

Final Exam Formulation
Math 5320

The final exam will consist of problems of the following types:

1. Problems and/or direct calculations illustrating the operations/arithmetic of $\mathbb{C}$
[Similar to problem 1 or 4 on Exam I]
2. Verification of a topological property of a metric space (a set A is open if condition XXX – a set A is closed if condition YYY – a point x is a limit point of a set A if condition ZZZ – a set K is compact if condition WWW, etc.)
[Similar to problem 2, 3 or 6 on Exam I]
3. Classification Problem (which of the following sets are open, closed, bounded, connected, compact, etc.)
[Similar to problem 8 on Exam I]
4. Find the radius of convergence of a power series
[Similar to problem 1 on Exam II]
5. Construct a conformal map from one region G to another H
[Similar to problem 4 or 6 on Exam II - TakeHome]
6. Determine the image of a specific region under a specific Möbius transformation
[Similar to problem 6 on Exam II or problem 5 on Exam II - TakeHome]
7. Use the Cauchy-Riemann equations to:
 1. Verify a function is or is not analytic on a region
 2. Verify a function is constant on a region
[Similar to problem 2 on Exam II]
 3. Construct a harmonic conjugate of a given harmonic function on a region
[Similar to problem 7 on Exam II - TakeHome]
8. Verify a (complex) functional identity

9. Evaluate line integrals:
 1. Using Theorem 4.1.9 for piece-wise smooth paths of integration
 2. Using the Fundamental Theorem of Calculus for Line Integrals
 3. Using Cauchy's Integral Formula #0 (and its corollaries)
 4. Using Cauchy's Theorem #0
 10. Apply Louisville's Theorem
 11. Apply the Identity Theorem
 12. Apply Cauchy's Estimate
 13. Apply the corollaries to Theorem 4.2.8
 14. State and prove certain theorems
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Problems representing Question Types 1 - 8 will constitute 40% of the exam

Problems representing Question Types 9 - 14 will constitute 60% of the exam